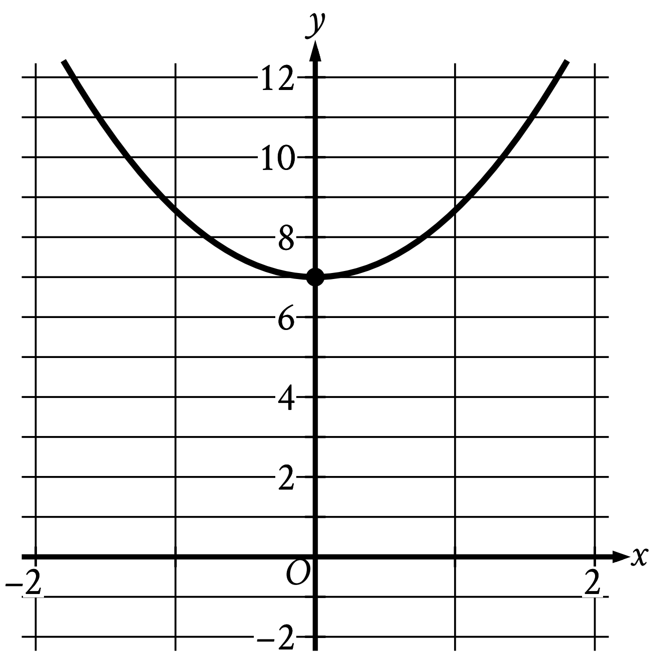
Question ID 2f4eafcc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■□□

ID: 2f4eafcc

PSAT2 M 3.1



The parabola shown intersects the *y*-axis at the point (x, y). What is the value of y?

ID: 2f4eafcc Answer

Correct Answer: 7

Rationale

The correct answer is 7. It's given that the parabola intersects the *y*-axis at the point x, y. The graph shows that the parabola intersects the *y*-axis at the point $0, 7$. Therefore, the value of y is 7.

Question Difficulty: Easy

Question ID 2e379126

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	■□□

ID: 2e379126

PSAT2 M 3.2

The function g is defined by $g(x) = 4x - 6$. What is the value of $g(-7)$?

A. -34

B. -22

C. $-\frac{13}{4}$

D. $-\frac{1}{4}$

ID: 2e379126 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the function g is defined by $gx = 4x - 6$. Substituting -7 for x into the given equation yields $g-7 = 4-7 - 6$, or $g-7 = -34$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of x for which $gx = -7$, not the value of $g-7$.

Question Difficulty: Easy

Question ID e006209c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	■□□

ID: e006209c

PSAT2 M 3.3

A geologist needs to collect at least 67 samples of lava from a volcano. If the geologist has already collected 63 samples from the volcano, what is the minimum number of additional samples the geologist needs to collect?

A. 130

B. 63

C. 4

D. 0

ID: e006209c Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the geologist has already collected 63 samples from the volcano. Let x represent the number of additional samples the geologist needs to collect. After collecting x additional samples, the geologist will have collected a total of $63 + x$ samples. It's given that the geologist needs to collect at least 67 samples. Therefore, $63 + x \geq 67$. Subtracting 63 from each side of this inequality yields the inequality $x \geq 4$. Thus, the geologist needs to collect a minimum of 4 additional samples.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of samples the geologist has already collected, rather than the minimum number of additional samples the geologist needs to collect.

Choice D is incorrect. If the geologist collects 0 additional samples, the geologist will have collected a total of 63 samples, which is less than 67 samples.

Question Difficulty: Easy

Question ID 0d43db90

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	■□□

ID: 0d43db90

PSAT2 M 3.4

The perimeter of triangle ABC is 17 inches, the length of side AB is 4 inches, and the length of side AC is 7 inches. What is the length, in inches, of side BC?

A. 4

B. 6

C. 7

D. 11

ID: 0d43db90 Answer

Correct Answer: B

Rationale

Choice B is correct. The perimeter of a triangle is the sum of the lengths of all three sides of the triangle. It's given that the lengths of side AB and side AC are 4 inches and 7 inches, respectively. Let x represent the length, in inches, of side BC. The sum of the lengths, in inches, of all three sides of triangle ABC can be represented by the expression $4 + 7 + x$. Since it's given that the perimeter of triangle ABC is 17 inches, it follows that $17 = 4 + 7 + x$, or $17 = 11 + x$. Subtracting 11 from both sides of this equation yields $6 = x$. Therefore, the length, in inches, of side BC is 6.

Choice A is incorrect. This is the length, in inches, of side AB. Choice C is incorrect. This is the length, in inches, of side AC.

Choice D is incorrect. This is the sum of the lengths, in inches, of sides AB and AC.

Question Difficulty: Easy

Question ID a130fcdc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	■□□

ID: a130fcdc

PSAT2 M 3.5

$g(x) = 11x + 4$

For the given linear function g, which table shows three values of x and their corresponding values of $g(x)$?

A.

x	$g(x)$
-1	7
0	11
1	15

B.

x	$g(x)$
-1	-4
0	0
1	4

C.

x	$g(x)$
-1	-7
0	4
1	15

D.

x	$g(x)$
-1	-11
0	0
1	11

ID: a130fcdc Answer

Correct Answer: C

Question ID c4259674

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■■□

ID: c4259674

PSAT2 M 3.6

The function f is defined by $f(x) = 4x^{-1}$. What is the value of $f(21)$?

A. -84

B. $\frac{1}{84}$

C. $\frac{4}{21}$

D. $\frac{21}{4}$

ID: c4259674 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that function f is defined by the equation $fx = 4x^{-1}$. The value of $f21$ is the value of fx when $x = 21$. Substituting 21 for x in the given equation yields $f21 = 421^{-1}$, which is equivalent to $f21 = 4\frac{1}{21}$, or $f21 = \frac{4}{21}$.

Choice A is incorrect. This is the value of $f21$ when $fx = -4x$, rather than $fx = 4x^{-1}$.

Choice B is incorrect. This is the value of $f21$ when $fx = 4x^{-1}$, rather than $fx = 4x^{-1}$.

Choice D is incorrect. This is the value of $f21$ when $fx = 4^{-1}x$, rather than $fx = 4x^{-1}$.

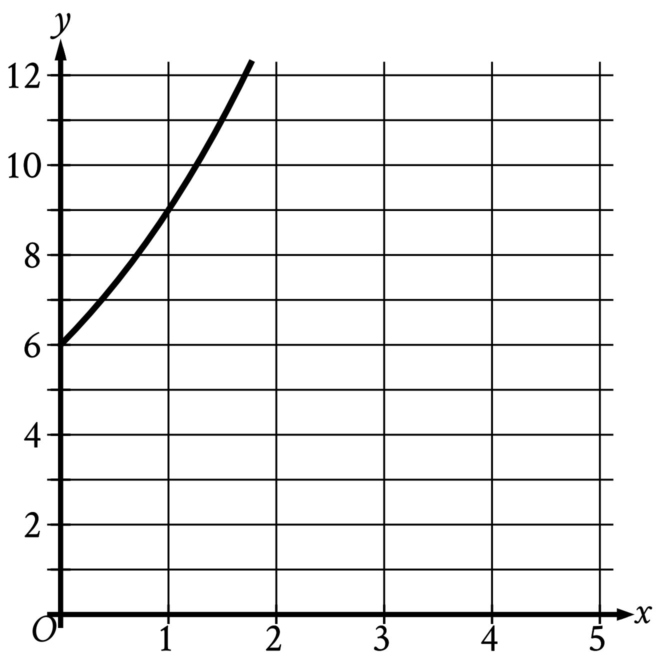
Question Difficulty: Medium

Question ID f1fa0821

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: f1fa0821

PSAT2 M 3.7



The graph gives the estimated population y, in thousands, of a town x years since 2003, where $0 \leq x \leq 5$. Which of the following best describes the increase in the estimated population from $x = 0$ to $x = 1$?

A. The estimated population at $x = 1$ is 0.5 times the estimated population at $x = 0$.

B. The estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.

C. The estimated population at $x = 1$ is 2.5 times the estimated population at $x = 0$.

D. The estimated population at $x = 1$ is 3.5 times the estimated population at $x = 0$.

ID: f1fa0821 Answer

Correct Answer: B

Rationale

Choice B is correct. On the graph shown, the *y*-axis represents estimated population, in thousands. The graph shows that when $x = 0$, the *y*-coordinate is 6. Therefore, the estimated population at $x = 0$ is 6 thousand. The graph also shows that when $x = 1$, the *y*-coordinate is 9. Therefore, the estimated population at $x = 1$ is 9 thousand. Dividing 9 thousand by 6 thousand yields 1.5; therefore, 9 thousand is 1.5 times 6 thousand. It follows that the estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

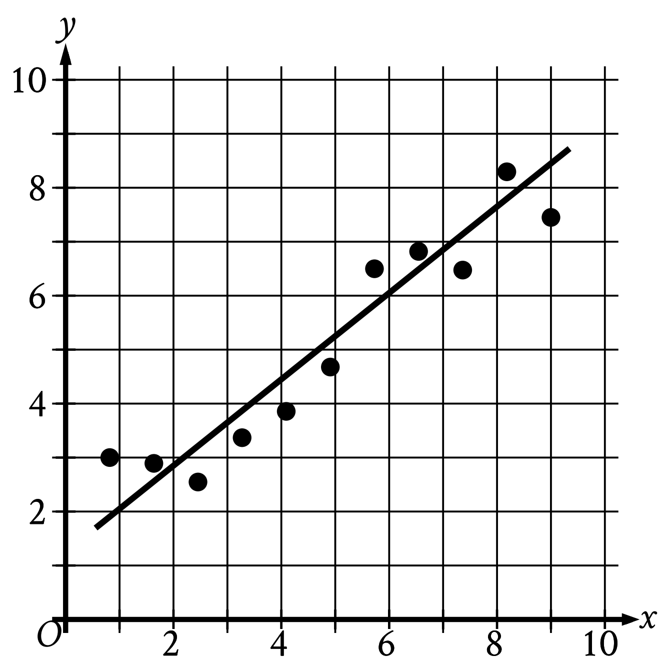
Question Difficulty: Medium

Question ID ad7dbb22

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	■■□

ID: ad7dbb22

PSAT2 M 3.8



The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown. For how many of the 11 data points does the line of best fit predict a greater y-value than the actual y-value?

ID: ad7dbb22 Answer

Correct Answer: 6

Rationale

The correct answer is 6. The line of best fit predicts a greater *y*-value than the actual *y*-value for any data point that's located below the line of best fit. For the scatterplot shown, 6 of the data points are below the line of best fit. Therefore, the line of best fit predicts a greater *y*-value than the actual *y*-value for 6 of the data points.

Question Difficulty: Medium

Question ID 55447be2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	■■□

ID: 55447be2

PSAT2 M 3.9

$y = \frac{1}{3}x - 14$ $y = -x + 18$ The solution to the given system of equations is (x, y). What is the value of x?

ID: 55447be2 Answer

Correct Answer: 24

Rationale

The correct answer is 24. The given system of equations can be solved by the substitution method. The first equation in the given system of equations is $y = \frac{1}{3}x - 14$. Substituting $\frac{1}{3}x - 14$ for y in the second equation in the given system yields $\frac{1}{3}x - 14 = -x + 18$. Adding 14 to both sides of this equation yields $\frac{1}{3}x = -x + 32$. Adding x to both sides of this equation yields $\frac{4}{3}x = 32$. Multiplying both sides of this equation by $\frac{3}{4}$ yields $x = 24$.

Question Difficulty: Medium

Question ID 8ad64841

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	■■□

ID: 8ad64841

PSAT2 M 3.10

A linear function f gives a company's profit, in dollars, for selling x items. The company's profit is $\$220$ when it sells 8 items, and its profit is $\$320$ when it sells 10 items. Which equation defines f?

A. $f(x) = 150x - 320$

B. $f(x) = 32x$

C. $f(x) = 50x - 10x$

D. $f(x) = 50x - 180$

ID: 8ad64841 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the relationship between x and fx is linear. A linear function can be written in the form $fx = mx + b$, where m is the slope and b is the *y*-coordinate of the *y*-intercept of the graph of $y = fx$ in the *xy*-plane. Given two points on a line, x_1, y_1 and x_2, y_2, the slope of the line can be found using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. It's given that the company's profit is $\$ 220$ when it sells 8 items and the profit is $\$ 320$ when it sells 10 items. Since fx represents the company's profit, in dollars, for selling x items, the graph of $y = fx$ in the *xy*-plane passes through the points $8, 220$ and $10, 320$. Substituting $8, 220$ and $10, 320$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{320 - 220}{10 - 8}$, which gives $m = \frac{100}{2}$, or $m = 50$. Substituting 50 for m, 8 for x, and 220 for fx in $fx = mx + b$ yields $220 = 508 + b$, or $220 = 400 + b$. Subtracting 400 from each side of this equation yields $-180 = b$. Substituting 50 for m and -180 for b in $fx = mx + b$ yields $fx = 50x - 180$. Therefore, the equation that defines f is $fx = 50x - 180$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 40049d49

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	■■□

ID: 40049d49

PSAT2 M 3.11

$4x + 12 = \frac{a(x+3)}{2}$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a?

A. 0

B. 3

C. 8

D. 12

ID: 40049d49 Answer

Correct Answer: C

Rationale

Choice C is correct. If an equation has infinitely many solutions, then the two sides of the equation must be equivalent. Multiplying each side of the given equation by 2 yields $8x + 24 = ax + 3$. Since 8 is a common factor of both terms on the left-hand side of this equation, the equation can be rewritten as $8x + 3 = ax + 3$. The two sides of this equation are equivalent when $a = 8$. Therefore, if the given equation has infinitely many solutions, the value of a is 8.

Alternate approach: If the given equation, $4x + 12 = \frac{ax + 3}{2}$, has infinitely many solutions, then both sides of this equation are equal for any value of x. If $x = 0$, then substituting 0 for x in the given equation yields $40 + 12 = \frac{a0 + 3}{2}$, or $12 = \frac{3}{2}a$. Dividing both sides of this equation by $\frac{3}{2}$ yields $8 = a$.

Choice A is incorrect. If the value of a is 0, the given equation is equivalent to $4x + 12 = 0$, which has one solution, not infinitely many solutions.

Choice B is incorrect. If the value of a is 3, the given equation is equivalent to $4x + 12 = \frac{3x + 3}{2}$, or $4x + 12 = \frac{3}{2}x + \frac{9}{2}$, which has one solution, not infinitely many solutions.

Choice D is incorrect. If the value of a is 12, the given equation is equivalent to $4x + 12 = \frac{12x + 3}{2}$, or $4x + 12 = 6x + 18$, which has one solution, not infinitely many solutions.

Question Difficulty: Medium

Question ID d72a2b4d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	■■□

ID: d72a2b4d

PSAT2 M 3.12

In March, the price of a collectible card was $\$15.50$. In April, the price of the collectible card was $\$17.36$. The price of the collectible card in April was $p\%$ of the price of the collectible card in March. What is the value of p?

A. 12

B. 88

C. 112

D. 188

ID: d72a2b4d Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the price of the collectible card was $\$15.50$ in March and $\$17.36$ in April. It's also given that the price of the collectible card in April was $p\%$ of the price in March. It follows that $\$17.36$ is $p\%$ of $\$15.50$. Therefore, the value of p can be calculated by solving the equation $17.36 = \frac{p}{100}15.50$, or $17.36 = \frac{15.50p}{100}$. Multiplying each side of this equation by 100 yields $1{,}736 = 15.50p$. Dividing each side of this equation by 15.50 yields $112 = p$. Therefore, the value of p is 112.

Choice A is incorrect. 12% is the percent increase in the price of the collectible card from March to April.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 3148fe3e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■■□

ID: 3148fe3e

PSAT2 M 3.13

$x^2 + y + 10 = 10$ $8x + 16 - y = 0$ The solution to the given system of equations is (x, y). What is the value of x?

A. -16

B. -4

C. 2

D. 8

ID: 3148fe3e Answer

Correct Answer: B

Rationale

Choice B is correct. Adding y to each side of the second equation in the given system of equations yields $8x + 16 = y$. Substituting $8x + 16$ for y in the first equation yields $x^2 + 8x + 16 + 10 = 10$. Subtracting 10 from each side of this equation yields $x^2 + 8x + 16 = 0$. This equation can be rewritten as $x + 4^2 = 0$. Taking the square root of each side of this equation yields $x + 4 = 0$. Subtracting 4 from each side of this equation yields $x =$ -4. Therefore, the value of x is -4.

Choice A is incorrect. This is the value of y, not x. Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID be1b8c74

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■■□

ID: be1b8c74

PSAT2 M 3.14

$x = 8a(b+9)$

The given equation relates the positive numbers a, b, and x. Which equation correctly expresses a in terms of b and x?

A. $a = \frac{x}{8} - (b+9)$

B. $a = \frac{x}{8(b+9)}$

C. $a = \frac{8(b+9)}{x}$

D. $a = 8x(b+9)$

ID: be1b8c74 Answer

Correct Answer: B

Rationale

Choice B is correct. To express a in terms of b and x, the given equation can be rewritten such that a is isolated on one side of the equation. Since it's given that b is a positive number, $b+9$ is not equal to zero. Therefore, dividing both sides of the given equation by $8b+9$ yields the equivalent equation $\frac{x}{8b+9} = a$, or $a = \frac{x}{8b+9}$.

Choice A is incorrect. This equation is equivalent to $x = 8a + b + 9$.

Choice C is incorrect. This equation is equivalent to $x = \frac{8b+9}{a}$. Choice D is incorrect. This equation is equivalent to $x = \frac{a}{8b+9}$.

Question Difficulty: Medium

Question ID fde6f3bb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	■■□

ID: fde6f3bb

PSAT2 M 3.15

$g(x) = \frac{3}{5}x + \frac{7}{6}$ $h(x) = 6x - 5$

The functions g and h are defined by the equations shown. Which expression is equivalent to $g(x) \cdot h(x)$?

A. $\frac{18x^2}{5} - \frac{35}{6}$

B. $\frac{18x^2}{5} + \frac{27x}{11} - \frac{35}{6}$

C. $\frac{18x^2}{5} - 4x - \frac{35}{6}$

D. $\frac{18x^2}{5} + 4x - \frac{35}{6}$

ID: fde6f3bb Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that $gx = \frac{3}{5}x + \frac{7}{6}$ and $hx = 6x - 5$. Substituting $\frac{3}{5}x + \frac{7}{6}$ for gx and $6x - 5$ for hx in the expression $gx \cdot hx$ yields $\frac{3}{5}x + \frac{7}{6}6x - 5$. This expression can be rewritten as $\frac{3}{5}x6x - 5 + \frac{7}{6}6x - 5$, or $\frac{18x^2}{5} - 3x + 7x - \frac{35}{6}$, which is equivalent to $\frac{18x^2}{5} + 4x - \frac{35}{6}$.

Choice A is incorrect. This expression is equivalent to $\frac{3}{5}x6x + \frac{7}{6}-5$, not $\frac{3}{5}x + \frac{7}{6}6x - 5$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This expression is equivalent to $\frac{3}{5}x - \frac{7}{6}6x + 5$, not $\frac{3}{5}x + \frac{7}{6}6x - 5$.

Question Difficulty: Medium

Question ID 301faf80

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■■■

ID: 301faf80

PSAT2 M 3.16

The product of two positive integers is 462. If the first integer is 5 greater than twice the second integer, what is the smaller of the two integers?

ID: 301faf80 Answer

Correct Answer: 14

Rationale

The correct answer is 14. Let x represent the first integer and y represent the second integer. If the first integer is 5 greater than twice the second integer, then $x = 2y + 5$. It's given that the product of the two integers is 462; therefore $xy = 462$. Substituting $2y + 5$ for x in this equation yields $(2y + 5)(y) = 462$, which can be written as $2y^2 + 5y = 462$. Subtracting 462 from each side of this equation yields $2y^2 + 5y - 462 = 0$. The left-hand side of this equation can be factored by finding two values whose product is $2(-462)$, or -924, and whose sum is 5. The two values whose product is -924 and whose sum is 5 are 33 and -28. Thus, the equation $2y^2 + 5y - 462 = 0$ can be rewritten as $2y^2 - 28y + 33y - 462 = 0$, which is equivalent to $2y(y - 14) + 33(y - 14) = 0$, or $(2y + 33)(y - 14) = 0$. By the zero product property, it follows that $2y + 33 = 0$ or $y - 14 = 0$. Subtracting 33 from both sides of the equation $2y + 33 = 0$ yields $2y = -33$. Dividing both sides of this equation by 2 yields $y = -\frac{33}{2}$. Since y is a positive integer, the value of y isn't $-\frac{33}{2}$. Adding 14 to both sides of the equation $y - 14 = 0$ yields $y = 14$. Substituting 14 for y in the equation $xy = 462$ yields $x(14) = 462$. Dividing both sides of this equation by 14 yields $x = 33$. Therefore, the two integers are 14 and 33, so the smaller of the two integers is 14.

Question Difficulty: Hard

Question ID 40491607

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 40491607

PSAT2 M 3.17

$f(x)=(x-1)(x+3)(x-2)$

In the *xy*-plane, when the graph of the function f, where $y=f(x)$, is shifted up 6 units, the resulting graph is defined by the function g. If the graph of $y=g(x)$ crosses through the point $(4,b)$, where b is a constant, what is the value of b?

ID: 40491607 Answer

Correct Answer: 48

Rationale

The correct answer is 48. It's given that in the *xy*-plane, when the graph of the function f, where $y = fx$, is shifted up 6 units, the resulting graph is defined by the function g. Therefore, function g can be defined by the equation $gx = fx + 6$. It's given that $fx = x - 1x + 3x - 2$. Substituting $x - 1x + 3x - 2$ for fx in the equation $gx = fx + 6$ yields $gx = x - 1x + 3x - 2 + 6$. For the point $4, b$, the value of x is 4. Substituting 4 for x in the equation $gx = x - 1x + 3x - 2 + 6$ yields $g4 = 4 - 14 + 34 - 2 + 6$, or $g4 = 48$. It follows that the graph of $y = gx$ crosses through the point $4, 48$. Therefore, the value of b is 48.

Question Difficulty: Hard

Question ID 50b99b2d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	■■■

ID: 50b99b2d

PSAT2 M 3.18

Objects R and S each travel at a constant speed. The speed of object R is half the speed of object S. Object R travels a distance of $4x$ inches in y seconds. Which expression represents the time, in seconds, it takes object S to travel a distance of $24x$ inches?

A. $12y$

B. $3y$

C. $16y$

D. $6y$

ID: 50b99b2d Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that object R travels a distance of $4x$ inches in y seconds. This speed can be written as $\frac{4x \text{ inches}}{y \text{ seconds}}$. It's given that the speed of object R is half the speed of object S. It follows that the speed of object S is twice the speed of object R, which is $2\frac{4x \text{ inches}}{y \text{ seconds}}$, or $\frac{8x \text{ inches}}{y \text{ seconds}}$. Let n represent the time, in seconds, it takes object S to travel a distance of $24x$ inches. The value of n can be found by solving the equation $\frac{8x \text{ inches}}{y \text{ seconds}} = \frac{24x \text{ inches}}{n \text{ seconds}}$, which can be written as $\frac{8x}{y} = \frac{24x}{n}$. Multiplying each side of this equation by ny yields $8xn = 24xy$. Dividing each side of this equation by $8x$ yields $n = 3y$. Therefore, the expression $3y$ represents the time, in seconds, it takes object S to travel a distance of $24x$ inches.

Choice A is incorrect. This expression represents the time, in seconds, it would take object S to travel a distance of $24x$ inches if the speed of object R were twice, not half, the speed of object S.

Choice C is incorrect. This expression represents the time, in seconds, it takes object S to travel a distance of $128x$ inches, not $24x$ inches.

Choice D is incorrect. This expression represents the time, in seconds, it takes object R, not object S, to travel a distance of $24x$ inches.

Question Difficulty: Hard

Question ID 7355b9d9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 7355b9d9

PSAT2 M 3.19

If $k - x$ is a factor of the expression $-x^2 + \frac{1}{29}nk^2$, where n and k are constants and $k > 0$, what is the value of n?

A. -29

B. $-\frac{1}{29}$

C. $\frac{1}{29}$

D. 29

ID: 7355b9d9 Answer

Correct Answer: D

Rationale

Choice D is correct. If $k - x$ is a factor of the expression $-x^2 + \frac{1}{29}nk^2$, then the expression can be written as $k - xax + b$, where a and b are constants. This expression can be rewritten as $akx + bk - ax^2 - bx$, or $-ax^2 + ak - bx + bk$. Since this expression is equivalent to $-x^2 + \frac{1}{29}nk^2$, it follows that $-a = -1$, $ak - b = 0$, and $bk = \frac{1}{29}nk^2$. Dividing each side of the equation $-a = -1$ by -1 yields $a = 1$. Substituting 1 for a in the equation $ak - b = 0$ yields $k - b = 0$. Adding b to each side of this equation yields $k = b$. Substituting k for b in the equation $bk = \frac{1}{29}nk^2$ yields $k^2 = \frac{1}{29}nk^2$. Since k is positive, dividing each side of this equation by k^2 yields $1 = \frac{1}{29}n$. Multiplying each side of this equation by 29 yields $29 = n$.

Alternate approach: The expression $x^2 - y^2$ can be written as $x - yx + y$, which is a difference of two squares. It follows that $\frac{1}{29}nk^2 - x^2$ is equivalent to $\sqrt{\frac{1}{29}n}\,k - x\sqrt{\frac{1}{29}n}\,k + x$. It's given that $k - x$ is a factor of $-x^2 + \frac{1}{29}nk^2$, so the factor $\sqrt{\frac{1}{29}n}\,k - x$ is equal to $k - x$. Adding x to both sides of the equation $\sqrt{\frac{1}{29}n}\,k - x = k - x$ yields $\sqrt{\frac{1}{29}n}\,k = k$. Since k is positive, dividing both sides of this equation by k yields $\sqrt{\frac{1}{29}n} = 1$. Squaring both sides of this equation yields $\frac{1}{29}n = 1$. Multiplying both sides of this equation by 29 yields $n = 29$.

Choice A is incorrect. This value of n gives the expression $-x^2 + \frac{1}{29}-29k^2$, or $-x^2 - k^2$. This expression doesn't have $k - x$ as a factor.

Choice B is incorrect. This value of n gives the expression $-x^2 + \frac{1}{29}-\frac{1}{29}k^2$, or $-x^2 + -\frac{1}{841}k^2$. This expression doesn't have $k - x$ as a factor.

Choice C is incorrect. This value of n gives the expression $-x^2 + \frac{1}{29}\frac{1}{29}k^2$, or $-x^2 + \frac{1}{841}k^2$. This expression doesn't have $k - x$ as a factor.

Question Difficulty: Hard

Question ID dcf63c94

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: dcf63c94

PSAT2 M 3.20

$f(x) = 272(2)^x$

The function f is defined by the given equation. If $h(x) = f(x - 4)$, which of the following equations defines function h?

A. $h(x) = 17(2)^x$

B. $h(x) = 68(2)^x$

C. $h(x) = 272(16)^x$

D. $h(x) = 272(8)^x$

ID: dcf63c94 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $f(x) = 272(2)^x$ and $h(x) = f(x - 4)$. Substituting $x - 4$ for x in $f(x) = 272(2)^x$ yields $fx - 4 = 2722^{x-4}$. Substituting hx for $fx - 4$ in this equation yields $hx = 2722^{x-4}$. Using the properties of exponents, the function $h(x) = 272(2)^{x-4}$ can be rewritten as $hx = \frac{2722^x}{2^4}$, which is equivalent to $hx = \frac{2722^x}{16}$, or $h(x) = 17(2)^x$. Therefore, of the given choices, an equation that defines function h is $h(x) = 17(2)^x$.

Choice B is incorrect. This equation defines function h if $h(x) = f(x - 2)$, not $h(x) = f(x - 4)$.

Choice C is incorrect. This equation defines function h if $h(x) = f(4x)$, not $h(x) = f(x - 4)$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID f731d88b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	■■■

ID: f731d88b

PSAT2 M 3.21

In convex pentagon $ABCDE$, segment AB is parallel to segment DE. The measure of angle B is 139 degrees, and the measure of angle D is 174 degrees. What is the measure, in degrees, of angle C?

ID: f731d88b Answer

Correct Answer: 47

Rationale

The correct answer is 47. It's given that the measure of angle B is 139 degrees. Therefore, the exterior angle formed by extending segment AB at point B has measure $180 - 139$, or 41, degrees. It's given that segment AB is parallel to segment DE. Extending segment BC at point C and extending segment DE at point D until the two segments intersect results in a transversal that intersects two parallel line segments. One of these intersection points is point B, and let the other intersection point be point X. Since segment AB is parallel to segment DE, alternate interior angles are congruent. Angle CXD and the exterior angle formed by extending segment AB at point B are alternate interior angles. Therefore, the measure of angle CXD is 41 degrees. It's given that the measure of angle D in pentagon $ABCDE$ is 174 degrees. Therefore, angle CDX has measure $180 - 174$, or 6, degrees. Since angle C in pentagon $ABCDE$ is an exterior angle of triangle CDX, it follows that the measure of angle C is the sum of the measures of angles CDX and CXD. Therefore, the measure, in degrees, of angle C is $6 + 41$, or 47.

Alternate approach: A line can be created that's perpendicular to segments AB and DE and passes through point C. Extending segments AB and DE at points B and D, respectively, until they intersect this line yields two right triangles. Let these intersection points be point X and point Y, and the two right triangles be triangle BXC and triangle DYC. It's given that the measure of angle B is 139 degrees. Therefore, angle CBX has measure $180 - 139$, or 41, degrees. Since the measure of angle CBX is 41 degrees and the measure of angle BXC is 90 degrees, it follows that the measure of angle XCB is $180 - 90 - 41$, or 49, degrees. It's given that the measure of angle D is 174 degrees. Therefore, angle YDC has measure $180 - 174$, or 6, degrees. Since the measure of angle YDC is 6 degrees and the measure of angle CYD is 90 degrees, it follows that the measure of angle DCY is $180 - 90 - 6$, or 84, degrees. Since angles XCB, DCY, and angle C in pentagon $ABCDE$ form segment XY, it follows that the sum of the measures of those angles is 180 degrees. Therefore, the measure, in degrees, of angle C is $180 - 49 - 84$, or 47.

Question Difficulty: Hard

Question ID 2c288148

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■■■

ID: 2c288148

PSAT2 M 3.22

$\sqrt{k-x} = 58 - x$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

ID: 2c288148 Answer

Correct Answer: 231

Rationale

The correct answer is 231. It's given that $\sqrt{k-x} = 58 - x$. Squaring both sides of this equation yields $k - x = (58 - x)^2$, which is equivalent to the given equation if $58 - x > 0$. It follows that if a solution to the equation $k - x = (58 - x)^2$ satisfies $58 - x > 0$, then it's also a solution to the given equation; if not, it's extraneous. The equation $k - x = (58 - x)^2$ can be rewritten as $k - x = 3{,}364 - 116x + x^2$. Adding x to both sides of this equation yields $k = x^2 - 115x + 3{,}364$. Subtracting k from both sides of this equation yields $0 = x^2 - 115x + (3{,}364 - k)$. The number of solutions to a quadratic equation in the form $0 = ax^2 + bx + c$, where a, b, and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. Substituting -115 for b, 1 for a, and $3{,}364 - k$ for c in $b^2 - 4ac$ yields $(-115)^2 - 4(1)(3{,}364 - k)$, or $4k - 231$. The equation $0 = x^2 - 115x + (3{,}364 - k)$ has exactly one real solution if the discriminant is equal to zero, or $4k - 231 = 0$. Subtracting 231 from both sides of this equation yields $4k = 231$. Dividing both sides of this equation by 4 yields $k = 57.75$. Therefore, if $k = 57.75$, then the equation $0 = x^2 - 115x + (3{,}364 - k)$ has exactly one real solution. Substituting 57.75 for k in this equation yields $0 = x^2 - 115x + (3{,}364 - 57.75)$, or $0 = x^2 - 115x + 3{,}306.25$, which is equivalent to $0 = (x - 57.5)^2$. Taking the square root of both sides of this equation yields $0 = x - 57.5$. Adding 57.5 to both sides of this equation yields $57.5 = x$. To check whether this solution satisfies $58 - x > 0$, the solution, 57.5, can be substituted for x in $58 - x > 0$, which yields $58 - 57.5 > 0$, or $0.5 > 0$. Since 0.5 is greater than 0, it follows that if $k = 57.75$, or $4k = 231$, then the given equation has exactly one real solution. If $4k < 231$, then the discriminant, $4k - 231$, is negative and the given equation has no solutions. Therefore, the minimum possible value of $4k$ is 231.

Question Difficulty: Hard